Calculus and Linear Algebra

Fundamental Theorems of Continuous Functions

Lecture 16

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Lecture outline

- Quick review: continuity on a closed interval.
- Intermediate Value Theorem.
- Bolzano's theorem and existence of zeros.
- Existence versus uniqueness.
- Bisection as a constructive application.
- Extreme Value Theorem (Weierstrass).
- Why closedness, boundedness and continuity matter.
- Applications to equations and economic optimization.

Main goal

Learn what continuity can guarantee **without explicitly solving** the problem.

Recall: continuity on $[a, b]$

A function is continuous on $[a, b]$ if

- it is continuous at every point of (a, b);
- it is right-continuous at a;
- it is left-continuous at b.

Why the closed interval matters

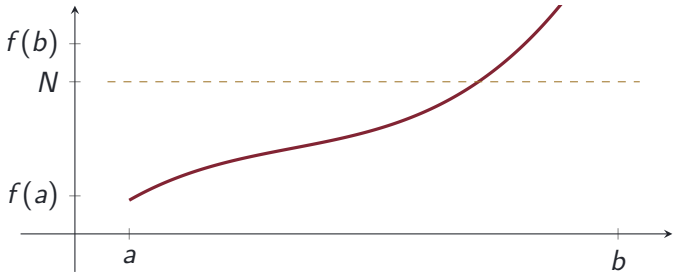
The endpoints belong to the domain, so the function is controlled on the entire interval, including its boundary.

Today we study two major consequences:

intermediate values and attainment of extrema.

Continuity prevents jumps

Suppose a continuous graph starts below a horizontal level and ends above it. It cannot cross from one side to the other without touching that level.



This geometric fact is the content of the Intermediate Value Theorem.

Intermediate Value Theorem

Theorem Let f be continuous on $[a, b]$. If N is any number between $f(a)$ and $f(b)$, then there exists at least one

$$c \in [a, b]$$

such that

$$\boxed{f(c) = N.}$$

In symbols, if for example

$$f(a) < N < f(b),$$

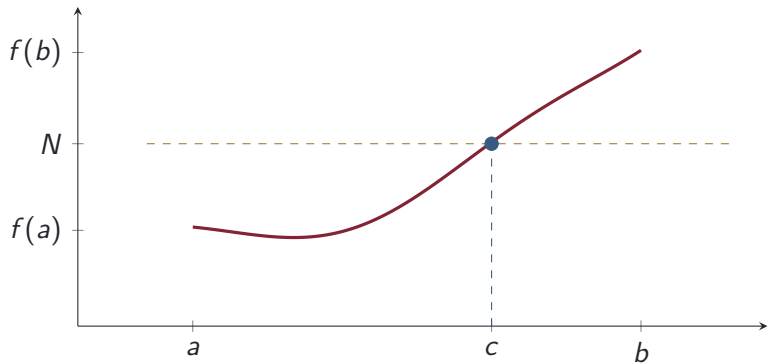
then

$$\exists\, c \in (a, b) : f(c) = N.$$

Interpretation

The image of an interval under a continuous function contains all intermediate values.

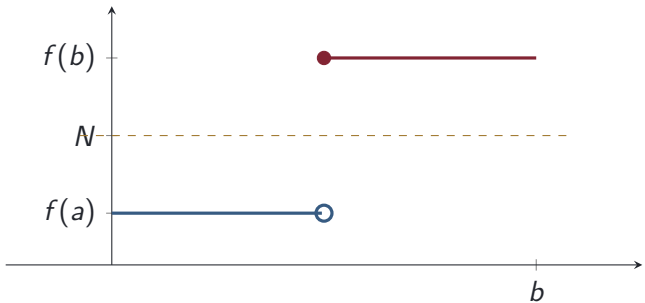
IVT: the graphical picture



The theorem guarantees at least one crossing, not necessarily exactly one.

Why continuity is essential

Without continuity, a function can jump over the target value.



Here N lies between the endpoint values, but the function never assumes N.

Proof idea for the IVT: bisection I

Assume

$$f(a) < N < f(b), \qquad g(x) = f(x) - N.$$

Then

$$g(a) < 0 < g(b).$$

Bisect $[a, b]$ and keep a half-interval whose endpoint values still have opposite signs. Repeat:

$$[a, b] \supset I_1 \supset I_2 \supset \cdots$$

with

$$\text{length}(I_n) = \frac{b-a}{2^n} \to 0.$$

Proof idea for the IVT: bisection II

The nested closed intervals shrink to a single point c. Choose endpoints a_n, b_n of I_n. Then

$$a_n \to c, \qquad b_n \to c,$$

and the construction gives

$$g(a_n) \leq 0 \leq g(b_n).$$

By continuity,

$$g(a_n) \to g(c), \qquad g(b_n) \to g(c).$$

Hence

$$\boxed{g(c) = 0,}$$

so $f(c) = N$.

Theoretical ingredient

This uses completeness of $\mathbb{R}$: nested closed intervals with shrinking lengths have a common point.

Example: reaching a prescribed value

Let

$$f(x) = x^3 + x + 1.$$

Show that f takes the value 5 somewhere in $[1, 2]$. Since f is a polynomial, it is continuous on $[1, 2]$.

$$f(1) = 3, \qquad f(2) = 11.$$

Because

$$3 < 5 < 11,$$

the IVT gives

$$\boxed{\exists\, c \in (1, 2) : f(c) = 5.}$$

No explicit solution of $x^3 + x + 1 = 5$ is required.

Bolzano's theorem: a zero exists

Theorem Let f be continuous on $[a, b]$. If

$$\boxed{f(a)f(b) < 0,}$$

then there exists at least one

$$c \in (a, b)$$

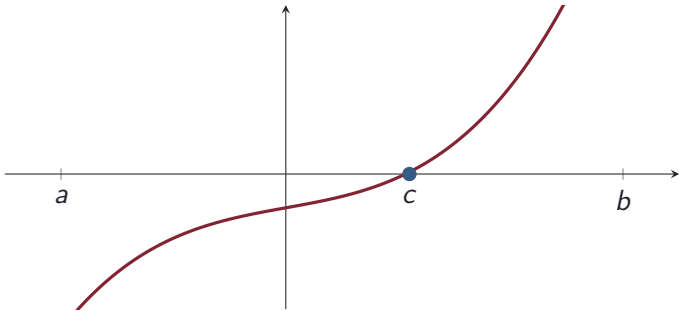
such that

$$\boxed{f(c) = 0.}$$

Why it follows from the IVT

If $f(a)$ and $f(b)$ have opposite signs, then 0 lies strictly between them. Apply the IVT with $N = 0$.

Bolzano: visual interpretation



Opposite signs at the endpoints force at least one crossing of the x-axis.

Exercise: prove that a root exists

Exercise Show that

$$x^3 + x - 1 = 0$$

has at least one solution in $(0, 1)$. Define

$$f(x) = x^3 + x - 1.$$

What must be checked before using Bolzano's theorem?

1. continuity on $[0, 1]$;
2. opposite signs at the endpoints.

Exercise: solution

Solution Since f is a polynomial, it is continuous on $[0,1]$.

$$f(0) = -1 < 0, \qquad f(1) = 1 > 0.$$

Therefore

$$f(0)f(1) < 0.$$

By Bolzano's theorem,

$$\boxed{\exists\, c \in (0,1) : c^3 + c - 1 = 0.}$$

The theorem proves existence, but does not yet give the exact root.

Existence is not uniqueness

Bolzano's theorem says

at least one zero.

It does not say

exactly one zero.

For example,

$$f(x) = x(x-1)(x+1)$$

is continuous and has three zeros:

$$-1, \quad 0, \quad 1.$$

How can uniqueness be obtained?

One common additional condition is **strict monotonicity**: a strictly monotone function can cross a horizontal level at most once.

Existence + strict monotonicity = uniqueness

Suppose f is continuous and strictly increasing on $[a, b]$, with

$$f(a) < 0 < f(b).$$

Bolzano gives at least one zero. Strict increase implies that two different points cannot have the same function value. Therefore the zero is unique:

$$\boxed{\exists!\, c \in (a, b) : f(c) = 0.}$$

For

$$f(x) = x^3 + x - 1,$$

we will later use derivatives to prove that f is strictly increasing.

Bisection method: from existence to approximation

Bolzano's theorem leads to a practical numerical method. If $f(a)f(b) < 0$:

1. compute the midpoint
$$m = \frac{a+b}{2};$$
2. evaluate $f(m)$;
3. keep the half-interval where the sign change remains;
4. repeat.

After n bisections, the interval length is

$$\boxed{\frac{b-a}{2^n}.}$$

Bisection example: $x^3 + x - 1 = 0$

Start with $[0, 1]$:

$$f(0) = -1, \qquad f(1) = 1.$$

First midpoint:

$$m_1 = 0.5, \qquad f(0.5) = 0.125 + 0.5 - 1 = -0.375.$$

The sign change is now on

$$[0.5, 1].$$

Second midpoint:

$$m_2 = 0.75, \qquad f(0.75) = 0.421875 + 0.75 - 1 > 0.$$

So the root lies in

$$\boxed{(0.5, 0.75).}$$

A common mistake

From

$$f(a)f(b) > 0$$

we cannot conclude that there is no zero in (a, b). Example:

$$f(x) = x^2 - 1$$

on $[-2, 2]$.

$$f(-2) = 3 > 0, \qquad f(2) = 3 > 0,$$

yet the function has two zeros:

$$-1, \quad 1.$$

Bolzano is a sufficient condition

A sign change guarantees a zero. No sign change does not rule zeros out.

From zeros to extrema

Continuity gives another fundamental guarantee. Suppose a continuous function is observed on a closed and bounded interval $[a, b]$. Can its values approach a maximum forever without ever reaching it? The Extreme Value Theorem says: **no**.

Extreme Value Theorem (Weierstrass)

Theorem If f is continuous on a closed and bounded interval $[a, b]$, then f attains both an absolute minimum and an absolute maximum. That is, there exist

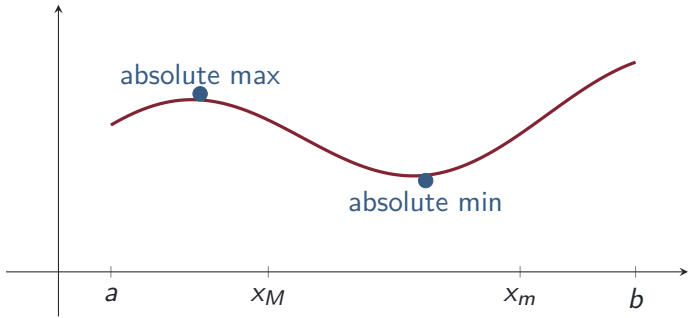
$x_m, x_M \in [a, b]$ such that

$$\boxed{f(x_m) \leq f(x) \leq f(x_M) \qquad \forall x \in [a, b].}$$

Equivalently,

$$\boxed{f(x_m) = \min_{x \in [a,b]} f(x), \qquad f(x_M) = \max_{x \in [a,b]} f(x).}$$

Weierstrass: graphical picture



The extrema may occur inside the interval or at its endpoints.

What the theorem guarantees – and what it does not

Weierstrass guarantees:

- a maximum value exists;
- a minimum value exists;
- both are actually attained by points in the interval.

It does **not** tell us:

- where the extrema occur;
- whether they are unique;
- how to calculate them efficiently.

Later

Derivatives will provide the main computational tool for locating candidate extrema.

Why the interval must be closed

Consider

$$f(x) = x$$

on the open interval

$$(0, 1).$$

The function is continuous, and its values can be arbitrarily close to 1. But

$$1 \notin (0, 1),$$

so no point in the domain satisfies $f(x) = 1$.

$$\boxed{\sup f = 1 \quad \text{but no maximum is attained.}}$$

The closedness hypothesis cannot simply be removed.

Why boundedness matters

Consider

$$f(x) = x$$

on

$$[0, +\infty).$$

The domain is closed, and f is continuous, but the interval is not bounded. There is no absolute maximum because

$$f(x) \to +\infty \qquad \text{as } x \to +\infty.$$

Thus a closed but unbounded interval is not enough for the theorem.

Why continuity matters

Define on $[0, 1]$

$$f(x) = \begin{cases} x, & 0 \le x < 1, \\ 0, & x = 1. \end{cases}$$

The domain is closed and bounded, but f is not continuous at 1. Values of $f(x)$ approach 1 as $x \to 1^-$, but the function never equals 1.

$$\boxed{\sup f = 1 \quad \text{but no maximum is attained.}}$$

Continuity is essential.

Proof idea for Weierstrass

A complete proof requires results about bounded sequences and compactness. The core idea is:

1. choose points $x_n \in [a, b]$ whose function values approach the supremum of f;
2. because $[a, b]$ is closed and bounded, a subsequence converges to some $x_M \in [a, b]$;
3. by continuity,
$$f(x_{n_k}) \to f(x_M);$$
4. since the left side approaches the supremum,
$$\boxed{f(x_M) = \sup f.}$$

The minimum follows similarly.

Example: existence of an optimum

Let

$$P(q) = -2q^2 + 12q - 5, \qquad q \in [0, 5],$$

represent profit as a function of output. P is a polynomial, hence continuous on the closed bounded interval $[0, 5]$. Therefore Weierstrass guarantees that

$$\boxed{P \text{ attains an absolute maximum and minimum on } [0, 5].}$$

At this stage we know the optimum exists. Later, derivatives will locate it.

Economic interpretation: why existence matters

In optimization problems we often write

$$\max_{q \in [a,b]} P(q).$$

Before differentiating, a basic theoretical question is:

Does an optimizer actually exist?

If the feasible set is a closed bounded interval and the objective is continuous, Weierstrass answers yes. This separates two questions:

1. **Existence**: does a maximizer/minimizer exist?
2. **Computation**: where is it?

Break-even as a zero of a continuous function

Let revenue and cost be continuous functions $R(q)$ and $C(q)$. Define profit

$$P(q) = R(q) - C(q).$$

If for two output levels $q_1 < q_2$,

$$P(q_1) < 0 < P(q_2),$$

then Bolzano gives

$$\boxed{\exists\, q^* \in (q_1, q_2) : P(q^*) = 0.}$$

Thus a break-even level exists between q_1 and q_2 even if we cannot solve $R(q) = C(q)$ explicitly.

Exercise: can Bolzano be applied?

Exercise Consider

$$f(x) = \frac{x+1}{x-1}$$

on $[0, 2]$. We have

$$f(0) = -1, \qquad f(2) = 3.$$

Can we conclude that $f(c) = 0$ for some $c \in (0, 2)$? Be careful: checking the endpoint signs is not enough.

Exercise: solution

Solution No. The function is not continuous on $[0, 2]$ because

$$x = 1$$

is not in its domain. Therefore the hypotheses of Bolzano's theorem fail. Indeed,

$$\frac{x+1}{x-1} = 0 \quad \Longleftrightarrow \quad x = -1,$$

which is outside $(0, 2)$.

Lesson

Always verify continuity on the entire interval before using an existence theorem.

Exercise: does a maximum exist?

Exercise For each case, decide whether Weierstrass guarantees an absolute maximum and minimum.

1. $f(x) = \ln x$ on $[1, 4]$.
2. $g(x) = \dfrac{1}{x}$ on $[-1, 1]$.
3. $h(x) = x^2$ on $(-2, 2)$.
4. $p(x) = e^{-x^2}$ on $[-3, 3]$.

For each case, identify exactly which hypothesis holds or fails.

Exercise: solution

Solution

1. $\ln x$ on $[1, 4]$: **yes**; continuous on a closed bounded interval.
2. $1/x$ on $[-1, 1]$: **no**; not continuous on the interval because $x = 0$ is excluded.
3. x^2 on $(-2, 2)$: **no guarantee**; the interval is not closed.
4. e^{-x^2} on $[-3, 3]$: **yes**; continuous on a closed bounded interval.

Do not confuse the three theorems

Theorem	**Conclusion**
Intermediate Value Theorem	Every value between $f(a)$ and $f(b)$ is attained.
Bolzano	Under a sign change, at least one zero is attained.
Weierstrass	Absolute maximum and minimum are attained on $[a, b]$.

All require continuity; Weierstrass also needs the interval to be closed and bounded.

A combined problem

Exercise Let

$$f(x) = x^3 - 4x + 1.$$

1. Explain why f is continuous on $[-3, 3]$.
2. Show that f has a zero in $(-3, -2)$.
3. Show that f has another zero in $(0, 1)$.
4. Does Weierstrass guarantee an absolute maximum and minimum on $[-3, 3]$?
5. Do the theorems tell us where those extrema occur?

Combined problem: solution outline

Solution f is a polynomial, hence continuous.

$$f(-3) = -14 < 0, \qquad f(-2) = 1 > 0,$$

so Bolzano gives a zero in $(-3, -2)$.

$$f(0) = 1 > 0, \qquad f(1) = -2 < 0,$$

so another zero lies in $(0, 1)$. Because $[-3, 3]$ is closed and bounded, Weierstrass guarantees absolute extrema. The theorem guarantees their existence, but not their locations.

Independent practice

Exercise

1. Prove that $x^5 + x - 2 = 0$ has a solution in $(1, 2)$.
2. Find an interval of length $1/4$ containing a root of $x^3 + x - 1 = 0$ by bisection.
3. Give an example of a continuous function on an open interval that has no maximum.
4. Explain why $f(x) = 1/(x - 1)$ on $[0, 2]$ does not satisfy Weierstrass.
5. A continuous profit function P satisfies $P(10) < 0 < P(20)$. What can you conclude?

The logical structure

Continuity on an interval

$\Downarrow$

Intermediate values

- IVT;
- Bolzano;
- existence of solutions;
- bisection.

Extrema on $[a, b]$

- Weierstrass;
- existence of max/min;
- optimization problems.

Looking ahead: from existence to local change

Continuity tells us that a graph has no jumps. But it does not measure how rapidly the graph changes. The next question is local:

$$\boxed{\text{What is the instantaneous rate of change of } f?}$$

In Lecture 17 we introduce the derivative as a limit of secant slopes:

$$\boxed{f'(x_0) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}.}$$

This will connect limits, geometry and economic marginal analysis.